

Percentages

Name: _____

IGCSE 0607 Extended Mathematics Practice Questions
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- 1 (a) Express 40 as a percentage of 200.

..... [1]

- (b) A bank offers simple interest at a rate of 5% per year. Calculate the interest earned on an investment of 600 AUD over 3 years.

..... [2]

- (c) A shop sells a coat for 90 HKD, which gives a 50% profit on the cost price. Calculate the cost price of the coat.

..... [2]

- 2** The mass of a breeding colony of insects increases by 10% each week. At the start of the first week, the mass is 500 grams.

(a) Calculate the mass of the colony at the end of 2 weeks.

..... [3]

(b) Express the mass at the end of 2 weeks as a percentage of the original mass.

..... [1]

- 3** An antique vase is purchased for 250 GBP.

(a) The value of the vase increases by 12% after one year. Calculate its new value.

..... [2]

(b) Due to a minor scratch, the new value then decreases by 10% . Calculate the final value of the vase.

..... [2]

- 4** A rare collector card is sold for 180 EUR, which represents a profit of 20% on the original purchase price. Calculate the original purchase price.

..... [3]

- 5** (a) A trader buys a bicycle for 150 AUD and sells it for 210 AUD. Calculate the percentage profit.

..... [2]

- (b) A population of bacteria is initially 5000. It increases by 20% in the first hour and then decreases by 10% in the second hour. Calculate the final population.

..... [2]

- (c) Express 150 HKD as a percentage of 60 HKD.

..... [2]

- 6** (a) Find the number of years it takes for 500 GBP to earn 75 GBP in simple interest at a rate of 3% per year.

..... [2]

- (b) A painting increases in value by 40% . It is now worth 2800 EUR. Calculate its original value.

..... [2]

- (c) A television is priced at 400 GBP. It is reduced by 15% in a clearance sale. An additional 10% discount is then applied to the sale price at the till. Calculate the final price of the television.

..... [2]

- 7 In a school, 60% of the students are girls.
The number of boys increases by 25% and the number of girls increases by 10%.
Find the overall percentage increase in the total number of students in the school.

..... [4]

- 8 A shopkeeper buys an item for x GBP. He marks it up by 50% to determine the selling price. During a clearance sale, he offers a 20% discount on this selling price.

(a) Show that the overall percentage profit is 20%.

[2]

(b) The profit made during the clearance sale is 25 GBP. Calculate the original cost price, x GBP.

..... [2]

- 9 Carlos invests P GBP at a rate of $r\%$ per year simple interest for 2 years. The total interest earned is 120 GBP.
He also invests the same amount P GBP at $r\%$ per year compound interest for 2 years. The total interest earned is 123 GBP.

(a) Show that $P \times r = 6000$.

[2]

(b) Calculate the value of r and the value of P .

..... [4]

 **10** An investor opens a savings account with an initial deposit of 4500 GBP. The account accrues simple interest at a fixed annual rate of 3.85% over a period of 6 years.

(a) Calculate the total amount of simple interest earned by the investor at the end of the 6-year period.

..... [2]

(b) Find the final value of the investment when the 6 years expire.

..... [1]

-  **11** A property investor buys a house for 452000 GBP. The value of the house increases by 4.2% each year.

(a) Calculate the value of the house after 5 years.

..... [3]

(b) The house sits on a rectangular plot of land. The length of the plot is increased by 15% and the width is decreased by 8% .

(i) Calculate the overall percentage change in the area of the plot.

..... [3]

(ii) Show that the new area is exactly 1.058 times the original area.

[2]

(c) The investor sells another property for 318500 GBP. This price includes a 12.5% capital gains tax. Calculate the price of the property before the tax was added.

..... [3]

-  **12** (a) A car is bought for 24 500 USD. Its value depreciates by 12.5% each year. Calculate the value of the car after 4 years. Give your answer correct to 3 significant figures.

..... [3]

- (b) During a sale, the price of a laptop is reduced by 15% to 845 USD. Calculate the original price of the laptop.

..... [3]

- (c) A sum of 8 000 USD is invested in an account paying 5.2% per year compound interest. Calculate the minimum number of complete years for the value of the investment to exceed 12 000 USD.

..... [3]

-  **13** (a) The annual number of tourists visiting a city was 1 450 000. The following year, the number was 1 310 000. Calculate the percentage decrease.

..... [2]

- (b) A travel agent sells a premium holiday package for 2 450 USD. This price includes a 25% profit on the agent's cost price. Calculate the cost price.

..... [3]

- (c) 9 600 USD is invested at 4.8% per year compound interest. Calculate the number of complete years required for the investment to double in value.

..... [3]